

Aditi Singh

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EDUCATION

- Indira Gandhi Delhi Technical University for Women** New Delhi, India
Bachelor of Technology in Electronics and Communications Engineering 2023-2027

SKILLS

- Languages:** C, C++, Python, VHDL
- Frameworks:** Tensorflow, Keras, OpenCV, PyTorch
- Hardware Platforms:** Arduino
- Non-technical Skills:** Content Writing, Event Management, UI/UX Design

EXPERIENCE

- ACM IGD TUW** Oct 2024 - Apr 2025
Research Intern
 - Co-authored a research paper titled “A Systematic Review on Automatic Legal Text Summarization Approaches”
 - Emphasized the need for automatic summarization of legal documents to improve accessibility for non-legal professionals
 - Reviewed a wide range of machine learning and deep learning approaches used in legal text summarization, including transformer-based models
 - Analyzed existing datasets such as BillSum and other domain-specific corpora used in prior research
 - Evaluated summarization methodologies, comparing extractive vs. abstractive techniques in terms of performance, complexity, and applicability
 - Studied the challenges posed by legal jargon, complex syntax, and rhetorical structure in achieving high-quality summarization
- Thinking Beyond** Sept 2024 - Dec 2024
AI Research Fellow
 - Completed the intensive Preparation Stage strengthening skills in Python programming, Linear Algebra, Calculus, and foundational AI concepts to develop mathematical models and implementing them numerically.
 - Conducted a five-week research project under academic mentorship, applied theoretical concepts to solve real-world Machine Learning challenges through a comprehensive study on the six popular optimization algorithms and evaluated on the basis of Accuracy, Precision, Recall and F1-score.
 - Developed a classifier model using CNN architecture to perform a detailed comparative analysis of the selected optimizers with different batch sizes ranging from 16 to 512.
 - Prepared a research poster and paper to compile the findings and methodology. Selected amongst **top 6 teams** to present the research poster at the Grand Finale of the first Beyond AI cohort.

RESEARCH AND PUBLICATIONS

- A. Singh, Y. Yifat, A Comparative Analysis of Optimizers for Classification with CNN (2024). *BeyondAI: Introduction to AI and Research Programme (4)*. Thinking Beyond. 

PROJECTS

- Object Sensing Smart Waste Bin**
Implemented object-sensing smart waste bin through UNO R3 SMD Board integrated with HC-SR04 ultrasonic sensor for precise distance measurement to detect waste objects. The system was programmed using the Arduino IDE to enable automated lid operation based on object proximity for efficient and hygienic waste disposal.
- Sound to Electricity Converter**
Developed a sound-to-electricity converter using UNO R3 SMD Board integrated with LM393 Sound Sensor Module for real-time sound detection and signal processing. Programmed the system using Arduino IDE in C to ensure efficient sound-to-electric signal conversion and accurate response to varying sound intensities.

HONORS AND AWARDS

- Recipient of **AWS AI-ML Scholarship** for **AI Programming with Python** Nanodegree program offered by Udacity, awarded to top 500 candidates globally 
- Recipient of **Bertelsmann Next Generation Tech Booster Scholarship** for **Foundation of Generative AI** Nanodegree program offered by Udacity 
- Honored twice to represent **India** at Harvard Project for International Relations as a **Delegate**
- Participant of **SAWIT.AI Learnathon** Program as a part of GUVI's 2nd **World Record Attempt** 
- Top 5** at IDeaGen UI/UX Hackathon -**prototype** 
- Awarded **Gold Medal** in Delhi State Yoga Championship and **Bronze Medal** in National Yoga Championship
- Served as **Head Girl** at CRPF Public School

VOLUNTEER EXPERIENCE

- Mentor, Education Mentoring Program, Govt of NCT Delhi:** Mentored 4 mentees on career options and sustainability, conducted awareness sessions.
- Operations Manager, Enactus IGD TUW:** Coordinated **3 field visits to NGOs**, enhancing outreach efforts and engaging with **250+** community members. Served as Operations Lead for the Young Barons Elevator Pitch 2024, successfully managing **1000+ participants**.
- Content Writer, AWS Cloud Club IGD TUW:** Developed comprehensive articles, tutorials and resources on **cloud computing** to contribute to educational content that enhanced the club's online presence and supported members to master AWS services and cloud concepts.